Fixed-degree isogenies between supersingular curves, and the rest of √élu: implementation and measurements

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Question. Are all algorithms in two sources implemented in `isogeny_algos`, and if not, implement them end to end?

1. Benčina, Kutas, Merz, Petit, Stopar and Weitkämper, *Finding isogenies of fixed degree between supersingular elliptic curves* (Isogeny Club, 12 December 2023; the slides were attached to the request).
2. Bernstein, De Feo, Leroux and Smith, *Faster computation of isogenies of large prime degree* (√élu, 2020; PDF attached).

Answer. Before this change, only √élu's core (Algorithm 2: the Weierstrass and Montgomery codomain and x-map) existed. Now every algorithm in both sources is implemented, tested against an independent computation, and run end to end, with these exceptions:

- the quantum speed-ups (Grover, Tani), for which there is only a query-count model;
- the rigorous Bauer–Joux step of the trivariate method, which is replaced by a run-time independence check.

All limits are listed in "Limits and negative results".

- **Status:** implemented and measured (this run), with the scope stated per item. Every claim below is either a test in the crate or a value in the run's records.
- **Run:** `run-20261016T0959Z`, commit `06e5deb47`; records in `data/records.jsonl` host in `data/host.txt`.
- **Host and conditions:** a 2-vCPU Xeon @ 2.8 GHz cloud container, shared with the build. One run per setting, so the wall times carry no interval (L0).

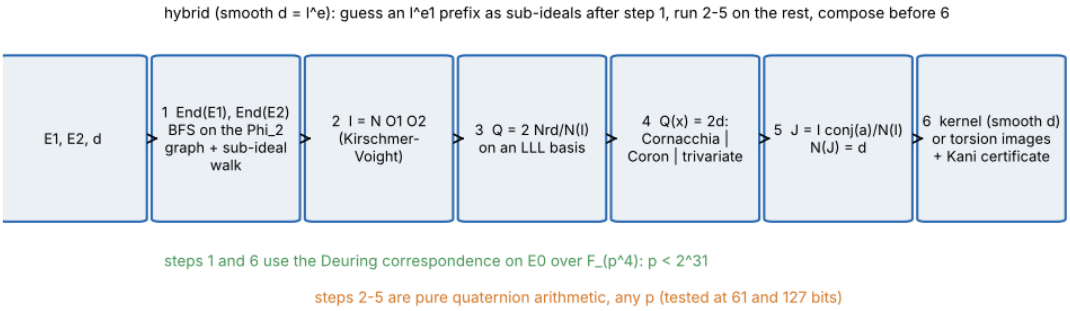


Figure 1: The six steps as implemented

The six steps of the slides as implemented. Steps 1 and 6 use the Deuring correspondence on E_0 over F_{p^4}, which limits p to below 2^{31}. Steps 2–5 are pure quaternion arithmetic and run at any size.

What was implemented

The code is in `isogeny_algos/src/`; the full status table is in `isogeny_algos/docs/SURVEY.md` §P1, P4 and P7 (the "V4" entries).

Benčina–Kutas–Merz–Petit–Stopar–Weitkämper (the slides' steps 1–6)

Step 1. End(E_1), End(E_2). In `fixed_degree/endring.rs`.

- *Algorithm:* a bidirectional BFS on the Φ_2 j-graph from 1728 to j(E), costing $\tilde{O}(\sqrt{p})$ vertices. The path is then walked on the quaternion side: the l + 1 sub-ideals $I \cdot l + I \cdot \beta$ ($\beta \in O_R(I)$) are exactly the l-isogenies, and the one whose Deuring image has the next j is kept. The ideal is reduced to minimal norm after each step. This is the classical form of the Eisenträger–Hallgren–Lauter–Morrison–Petit reduction.
- *Independent check:* the Deuring image of the result is j(E). The l + 1 sub-ideals' images are exactly the roots of $\Phi_l(j, Y)$, checked for l = 2 and 3.

Step 2. Connecting ideal. In `fixed_degree/connecting.rs`.

- *Algorithm:* Kirschmer–Voight, $I = N \cdot O_1 O_2$.
- *Independent check:* $O_L(I) = O_1$ and $O_R(I) = O_2$. It also matches $\mathrm{conj}(I_1) \cdot I_2$, and the Deuring image of $I_1 I$ is E_2.

Step 3. Norm form. In `connecting::norm_form`.

- *Algorithm:* $Q = 2 \cdot \mathrm{Nrd}/N(I)$ on an LLL-reduced basis, so the Gram matrix is integral.
- *Independent check:* $Q(x) \cdot N(I) = 2 \cdot \mathrm{Nrd}(\Sigma\, x_i \sigma_i)$ for random x.

Step 4a. Cornacchia. In `fixed_degree/cornacchia.rs` and `factor.rs`.

- *Algorithm:*
  - guess (x_3, x_4);
  - complete the square to $X^2 + |D|Y^2 = N$;
  - factor N, with Pollard rho;
  - take every $\sqrt{-D} \bmod N$ by CRT;
  - abandon the guess when $\omega(N) > 11 \cdot \log\log N$.

- *Independent check:* every solution set equals brute force. This covers $x^2 + dy^2 = m$ for all $m \le 2500$ and 12 values of d, and 300 random binary quadratic equations.

Step 4b. Coron's bivariate method. In `fixed_degree/coppersmith.rs`, with an exact integer LLL in `lattice.rs` and multivariate integer polynomials and Bareiss resultants in `mpoly.rs`.

- *Algorithm:* a box lattice; a vector is accepted only with the certificate $\|h(xX)\|_1 < n$; then Res_y, and the integer roots are found exactly.
- *Independent check:* planted roots are recovered up to Coron's bound (reached at $k = 2$).

Step 4c. Trivariate method (Bauer–Joux setting). In `fixed_degree/coppersmith.rs`. This one is **partial**.

- *Algorithm:* a Coron-style trivariate lattice. Instead of Bauer–Joux's rigorous Gröbner-basis step, the algebraic independence of the two polynomials is checked at run time by resultants modulo primes.
- *Independent check:* planted roots, but only for $|x_i| \lesssim 4$ (30-bit Gram entries) to $\lesssim 12$ (40-bit).

Step 5. The ideal of norm d. In `pipeline::solve_degree`.

- *Algorithm:* $J = I \cdot \mathrm{conj}(\alpha)/N(I)$.
- *Independent check:* $N(J) = d$, $O_L(J) = O_1$, and $O_R(J) = \beta^{-1} O_2 \beta$ exactly.

Step 6a. Ideal → isogeny, smooth d. In `fixed_degree/represent.rs`.

- *Algorithm:* the kernel $E_0[K + d \cdot O_0]$ is pushed through $E_0 \to E_1'$ and turned into a Vélu chain.
- *Independent check:* the chain from the caller's own E_1 lands on $j(E_2)$.

Step 6b. Ideal → isogeny, any odd d. Same file.

- *Algorithm:* compute $\chi(S_1)$, $\chi(S_2)$ on $E_1[2^{(a+2)}]$, the Robert-style representation, through $\varphi_K = [N(K')^{-1}] \cdot \varphi_{K'} \circ \theta$.
- *Certificate (Kani):* take an auxiliary isogeny of degree $2^a - d$, from a second random ideal and evaluated the same way. The $(2^a, 2^a)$-isogeny of $X \times E_2$ with kernel $\{(g(S), \chi(S))\}$ must split, and a twisted control kernel must not.
- *Independent check:* the Kani split, the twisted control, $e(\chi S_1, \chi S_2) = e(S_1, S_2)^d$, and the Deuring j.

Step 7. Hybrid. In `fixed_degree/hybrid.rs`.

- *Algorithm:* guess an l^{e_1} prefix as a chain of sub-ideals, solve the l^{e-e_1} rest, and compose.
- *Independent check:* the composite lands on E_2, and the degree-3^7 isogeny is Kani-certified.

Baselines. In `fixed_degree/baselines.rs`.

- *Algorithms:* exhaustive search for prime d (Φ_d roots, or kernels); MITM for $d = l^e$; van Oorschot–Wiener claw search.

- *Independent check:* planted paths are found and are cyclic. The counts are compared with theory in the MITM/vOW figure.

Quantum (Grover O(√d), Tani d^{1/3}). Only a query-count model in `baselines::grover_tani_cost_model`, because no quantum computer is available.

√élu, the parts that were missing

§2 Strassen. In `strassen.rs`.

- *Algorithm:* $\ell!$ mod n for any $n < 2^{62}$, and deterministic factoring from gcd(n, $\ell!$ mod n).
- *Independent check:* matches the naive product on 528 (ℓ, n) pairs; Wilson's theorem at $p \approx 2^{20}$; factoring matches `factor_u64`.

§3 Algorithm 1. In `kernel/sqrt_mult.rs`.

- *Algorithm:* the multiplicative h_S over I + J ∪ K.
- *Independent check:* equals the direct product over Zp, FpM<2> and FpM<8>; h_S = Φ_n for 11 values of n.

§4.14 Irrational generators. In `kernel/sqrt_velu_irr.rs`.

- *Algorithm:* index systems made of Frobenius orbits; E_J with coefficients in F_q; norms taken down to F_q.
- *Independent check:* equals Montgomery Vélu over F_{q^e} for e = 2, 3, 4.

§6 CSURF. In `path/csurf.rs`.

- *Algorithm:* radical 2-steps (the formula is derived in the module) and odd steps by Vélu or √élu.
- *Independent check:* every step matches Vélu and the modular polynomial Φ; the actions commute and invert at 62 bits, 111 bits and CSURF-512.

App. A.5. In `kernel/sqrt_velu_mont.rs`.

- *Algorithm:* fused Q(±1), Q(α), Q(1/α), and remainder trees with precomputed reciprocals.
- *Independent check:* equals Vélu on CSIDH-512 and at $\ell = 100003$.

Tests. `cargo test --release` in `isogeny_algos` passes 178 tests. The new test files are `tests/fixed_degree_pipeline.rs` (9 tests), `tests/fixed_degree_solvers.rs` (10), `tests/coppersmith.rs` (6) and `tests/sqrt_velu_gaps.rs` (15), plus 8 new unit tests.

Measurements

The harness is `isogeny_algos/examples/fixed_degree_report.rs` (native Rust). `plot.py` only lays out the records.

End-to-end pipeline, $p = 1605631 = 2^{15} \cdot 49 - 1$

These are 11 instances (figure "Wall time per step"). In every instance E_1 is the endpoint of an independent random 2-walk from 1728.

Planted prime d with $\sqrt{p} < d < 2^a = 4096$, Cornacchia (6 instances):

- All 6 were found.
- The torsion representation was **Kani-certified** in 6 of 6, and the pairing and Deuring-j checks passed in 6 of 6.

- d = 1399 … 3671, so log_p d = 0.507 … 0.574.
- Total time 1.9 – 3.9 s per instance; 94–99 % of it is step 1 (83–87 % in the random-target instances below).

Random E_2, nothing planted, prime d ≈ 8–12p, so log_p d ≈ 1.15–1.17 (3 instances):

- All 3 were found: x = [111, 113, 14, 9], [−35, 249, 10, −4], [185, −65, 10, 8].
- The norm equation took 226, 307 and 428 ms.
- The pairing and Deuring-j checks passed. Because d > 2^a there is no Kani certificate.

Coron's bivariate method at this small p (2 instances): **no solution in the search box**. On the same instances Cornacchia found x = [0, −1, 0, 1] and [−1, 0, 1, −1]. At this size the Gram entries are about $2\sqrt{p} \approx 2^{11}$, so W is too small for the lattice to certify any vector. See Limits.

The unit tests separately cover:

- planted smooth d (3·13·37 and 5·7·37 at p = 1477631), producing a kernel representation;
- the hybrid method with d = 3^7;
- one Coron success at p = 1605631 (`pipeline_coppersmith_bivariate`, where x = [−1, −1, 0, 0]).

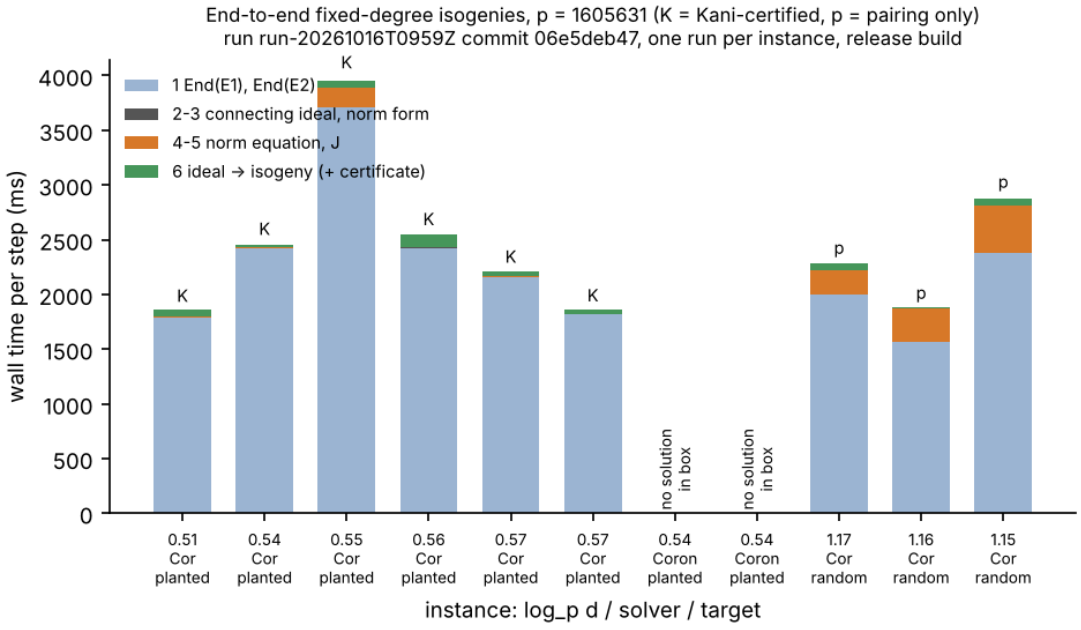


Figure 2: Wall time per step per pipeline instance

Wall time of each step for each pipeline instance (run-20261016T0959Z, one run each). K = Kani-certified torsion representation; p = checked by pairing and Deuring j only.

Steps 2–5 at cryptographic size

Each instance takes a random $I_1 \subset O_0$ and a planted J_0 of prime norm d. O_2 is the right order of a reduced ideal equivalent to $I_1 \cdot J_0$, so the Kirschmer–Voight ideal does not reveal d. There are two instances per setting (figure "Steps 2–5 wall time").

$p = 2^{61} - 1$. The table counts solved instances out of 2.

d/√p	log_p d	Cornacchia	Coron	trivariate
≈ 2–9	0.516–0.552	—	—	2/6 overall: at d/√p = 2 and 4; failed at 3, 5, 9, 9
≈ 6	0.542	2/2	2/2	—
≈ 20	0.57	2/2	2/2	—
≈ 90	0.606	2/2	2/2	—
≈ 380	0.64	2/2	2/2	—

Wall time at log_p d ≈ 0.64:

- Cornacchia 26.6 s and 71.2 s;
- Coron 6.5 s and 0.95 s.

Coron is faster than Cornacchia in 7 of the 8 (setting, repetition) pairs; each solver ran on its own random instances, so these are not paired on the same instance. The exception is rep 1 at d/√p ≈ 6 (0.019 s Cornacchia, 0.030 s Coron).

$p = 2^{127} - 1$, Coron only. It solved 6 of 6 for d/√p = 5–102 (log_p d 0.518–0.553), taking 0.04–4.3 s.

Cornacchia is not run at 127 bits. Every guess needs an N of about 2^{200} factored, which Pollard rho cannot do; the slides' L(1/3) cost assumes the number field sieve.

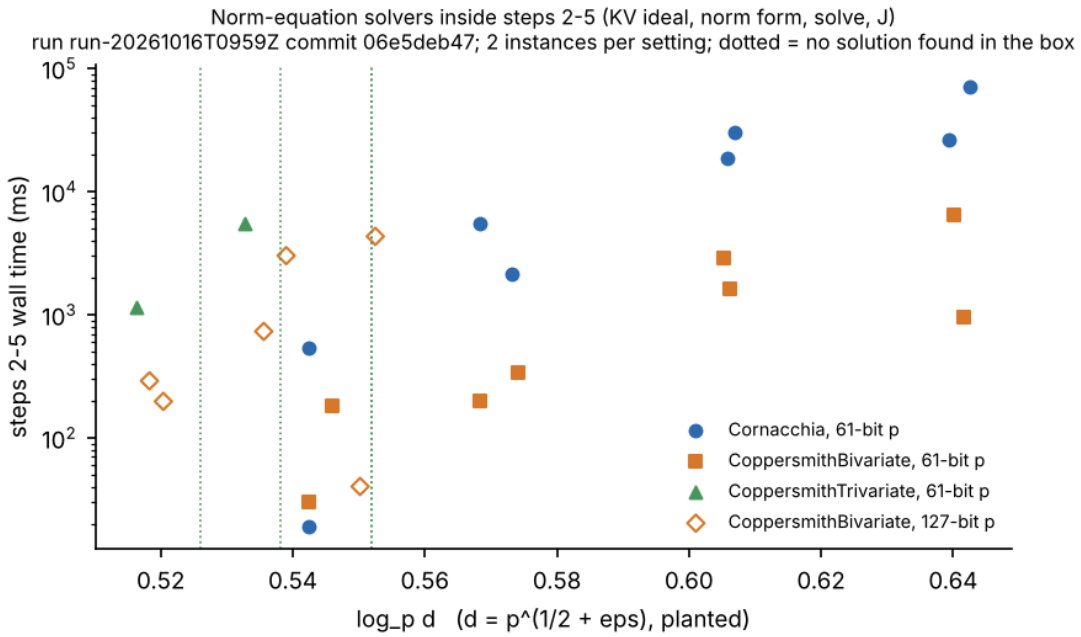


Figure 3: Steps 2–5 wall time against log_p d

Steps 2–5 wall time against log_p d. Filled markers are 61-bit p, open diamonds 127-bit p. Dotted lines mark instances where the trivariate method found no solution.

√élu §2: Strassen's modular factorial

$\ell!$ mod $(2^{61} - 1)$, mean over the repetitions stated in the records:

ℓ	naive (ms)	Strassen (ms)	Strassen / naive
10^3	0.044	0.046	1.05
10^4	0.42	0.43	1.02
10^5	4.06	1.72	0.42
10^6	45.1	11.7	0.26
$4{\cdot}10^6$	164.8	40.2	0.24

- Below b = ⌊√ℓ⌋ = 200, the code uses the naive product by design.
- The slope comes from Karatsuba multiplication: O(ℓ^0.79), not the paper's Õ(√ℓ), which needs quasi-linear polynomial multiplication.

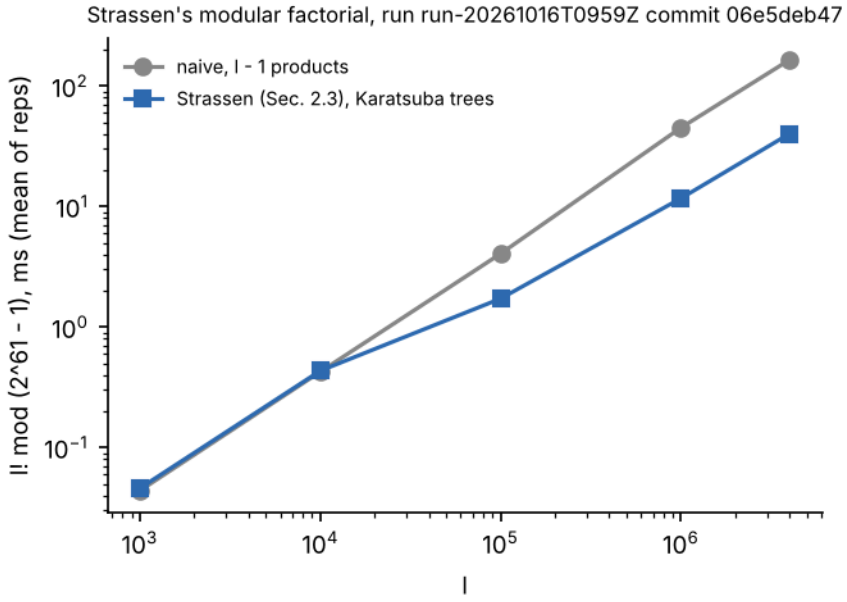


Figure 4: Strassen vs naive modular factorial

Baselines for d = 2^e (p = 4000039)

MITM. Leaves equal 2·3·2^{e/2−1} exactly for e = 12, 14, 16, 18 (figure "MITM and vOW counts").

vOW (w = 16) f-evaluations, against $2.5{\cdot}\sqrt{(|S|^3/w)}$:

e	rep 0	rep 1	expected
12	1030	1953	1663
14	1056	820	4703
16	5933	459	13302
18	64744	1016	37624

- The single-instance variance is large: the golden collision is found early or late.
- These spaces ($|S| \le 1536$) are too small to test the $1/\sqrt{w}$ scaling.

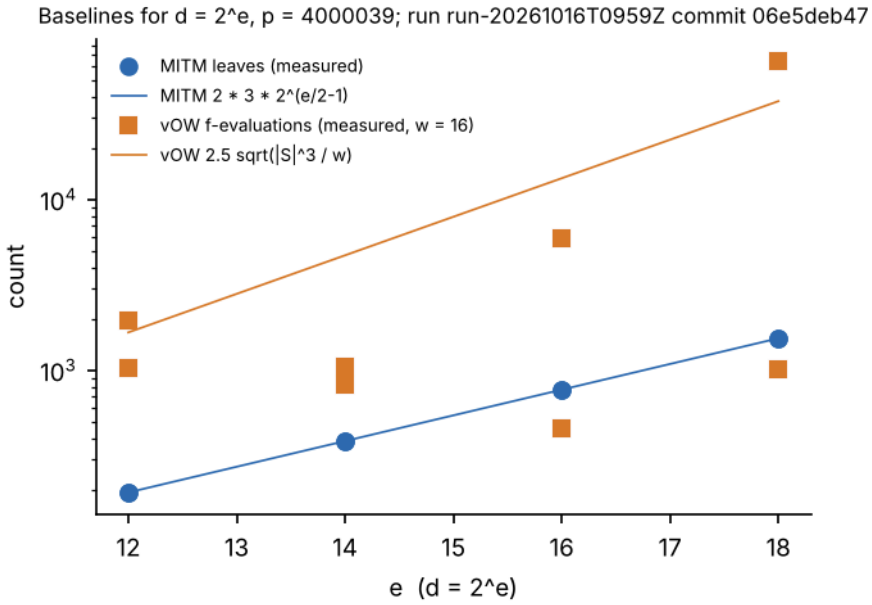


Figure 5: MITM and vOW counts against theory

CSURF-512

One action with key entries in $\{-1, 0, 1\}$ took 25.8 ms, and the inverse key returned to the starting curve.

Limits and negative results

1. **Prime size of the full pipeline.** The Deuring step (steps 1 and 6) is limited to $p < 2^{31}$ by `ext::Fp4`'s u128 arithmetic. It also needs a 256-smooth odd part of $p^2 - 1$ that is much larger than p; the primes were chosen by search. Steps 2–5 have no size limit.
2. **Kani certificate.** It needs $d < 2^a$, where 2^(a+3) | p + 1. In this run that means d < 4096 at p = 1605631, i.e. log_p d < 0.58. For larger d the torsion images are checked only by the Weil pairing and the Deuring j.
3. **Textbook auxiliaries do not work here (negative).** The usual auxiliary for the certificate is an endomorphism $\gamma \in \mathrm{End}(E_1)$ of degree $2^a - d$. In this regime it does not exist: $2^a - d < \sqrt{p}$, and only integers have reduced norm that small in a maximal order. Robert's dimension-4 integer matrix [[u, −w], [w, u]] was tried and failed: its first (2,2,2,2)-step is a product of two gluings, which `theta_g::chain_g` cannot start from. Integer α fails and Gaussian α works, checked on the crate's own Kani-4 instances. The certificate actually used is a second quaternion-computed isogeny of degree $2^a - d$. Kani's lemma needs only coprime degrees summing to 2^a.
4. **Coron needs large coefficients.** It failed in 2 of 2 pipeline instances at $p \approx 2^{21}$ where Cornacchia succeeded, while it worked in every 61- and 127-bit setting.

5. **The trivariate method is weak and heuristic.** It succeeded in 2 of 6 settings at 61 bits ($d/\sqrt{p} \le 4$) and in the test at $d/\sqrt{p} \approx 4$. The rigorous Bauer–Joux step is not implemented.
6. **Cornacchia's factoring.** It uses Pollard rho only, so N above about 2^{100} with two large factors gets abandoned.
7. **Strassen's complexity.** It is Karatsuba-limited, as measured above.
8. **Index systems.** §4.14 index systems for $e \ge 3$, and Algorithm 1's greedy index systems for unit groups with many factors, leave a large K. At the sizes tested, `SqrtVeluIrr` is slower than Vélu over the extension field, as the paper itself warns.
9. **CSURF-512 prime.** It was reconstructed from the paper's description. The paper's brace says "6 consecutive primes" but the list has 5; readings that add 359 or 397 are not prime. The prime was not checked against the authors' code.
10. **Quantum algorithms.** Not simulated.
11. **Lean formalisation.** There is none. The derived CSURF radical 2-step formula and the Kani-certificate argument are not formalised. The missing pieces are a Mathlib formalisation of Vélu's formulas and of Kani's lemma; the code's correctness rests on the tests listed above. All other items are implementations of published algorithms.

Reproduce

```
cd isogeny_algos
cargo test --release --test fixed_degree_pipeline --test fixed_degree_solvers --test
coppersmith --test sqrt_velu_gaps
cargo run --release --example fixed_degree_report > ../research/
fixed_degree_isogenies_20261016/data/records.jsonl
cd ../research/fixed_degree_isogenies_20261016 && python3 plot.py
FD_TRACE=1 cargo test --release --test fixed_degree_pipeline -- --nocapture   # per-
step trace
```

References

- B. Benčina, P. Kutas, S.-P. Merz, C. Petit, M. Stopar, C. Weitkämper, *Finding isogenies of fixed degree between supersingular elliptic curves*, Isogeny Club talk, 12 December 2023 (slides attached to the request).
- D. J. Bernstein, L. De Feo, A. Leroux, B. Smith, *Faster computation of isogenies of large prime degree*, 2020 (PDF attached to the request).
- J.-S. Coron, *Finding small roots of bivariate integer polynomial equations: a direct approach*, CRYPTO 2007.
- A. Bauer, A. Joux, *Toward a rigorous variation of Coppersmith's algorithm on three variables*, EUROCRYPT 2007.
- M. Kirschmer, J. Voight, *Algorithmic enumeration of ideal classes for quaternion orders*, SIAM J. Comput. 2010.
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- D. Kohel, K. Lauter, C. Petit, J.-P. Tignol, *On the quaternion ℓ-isogeny path problem*, LMS J. Comput. Math. 2014.
- D. Robert, *Evaluating isogenies in polylogarithmic time*, 2022.
- W. Castryck, T. Decru, *CSIDH on the surface*, PQCrypto 2020.

- G. Cornacchia, *Su di un metodo per la risoluzione in numeri interi dell'equazione* $\sum c_h x^{n-h} y^h = P$, 1908.

These references are cited from memory, except the two attached sources, which were read in this session.